



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO SREE SAPTHAGIRI AGENCIES

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.

2. Typical Weather Conditions and Potential Weather-Related Hazards

This region of Tamil Nadu experiences a tropical climate with hot summers, substantial monsoon rainfall between June and September, and mild winters. During monsoon season, heavy rainfall can cause waterlogging and reduce road visibility, presenting potential hazards for drivers.

3. Traffic Patterns

- **Peak Hours:** Traffic tends to be busier during morning (7:30 AM to 9:30 AM) and evening (6:00 PM to 8:00 PM) as local residents commute.
- **Congestion-Prone Areas:** Areas near market zones and schools can experience congestion, especially in Sangagiri and key points like Bhavani Main Road.

4. Assessment of Road Quality and Infrastructure

- **Road Quality:** The roads are generally in fair condition but may have potholes and uneven surfaces, particularly after the rainy season.
- **Infrastructure:** Limited presence of dedicated lanes for heavy vehicles; be cautious of abrupt turns and narrow crossings.

5. Suggestions for Alternative Routes

In case of emergencies or roadblocks, drivers can consider using the main highways like the Salem-Kochi Highway, although this might increase the distance and travel time.

6. Summary of Local Regulations for Hazardous Material Transport

- Strict adherence to transportation permits is required.
- Movement of hazardous materials is usually restricted during peak traffic hours and within school vicinity zones.
- Ensure all vehicles carrying hazardous materials have appropriate placards and spill kits.

7. Overview of Historical Incidents

While specific historical data may be limited, there have been past reports of vehicular accidents involving heavy vehicles, mainly due to poor road conditions and inclement weather during the monsoon.

8. Environmental Considerations and Sensitive Areas

- **Sensitive Areas:** Residential areas and agricultural fields are prevalent along the route.
- Minimize noise and emissions, and be conscious of local wildlife, particularly in rural zones.

9. Communication Coverage

- Generally good coverage is available along the route, but expect potential dead zones in more rural areas. Ensure drivers have pre-downloaded maps and emergency contacts.

10. Estimated Emergency Response Times

Response times can vary but are generally expected to be within 20-30 minutes in urban areas. In more rural segments, it might extend beyond 45 minutes.

11. Summary of Risk Assessment

Overall, the route presents moderate risks mainly due to weather conditions, road quality, and occasional congestion. Drivers should be vigilant during monsoon season and peak traffic hours. Proper communication devices and emergency protocols are essential for safety. Regular updates on road conditions and traffic advisories can further mitigate potential risks.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43811, 77.87348	15 KM/Hr
1	Turn	Medium	11.43956, 77.87340	30 KM/Hr
2	Turn	Medium	11.43971, 77.87348	30 KM/Hr
3	Turn	High	11.44029, 77.87544	15 KM/Hr
4	Turn	Medium	11.44898, 77.87410	30 KM/Hr
5	Turn	Medium	11.45348, 77.85706	30 KM/Hr
6	Turn	Medium	11.46318, 77.84913	30 KM/Hr
7	Turn	Medium	11.45697, 77.82719	30 KM/Hr
8	Turn	High	11.45542, 77.81725	15 KM/Hr
9	Turn	High	11.45631, 77.81668	15 KM/Hr
10	Turn	Medium	11.49249, 77.85821	30 KM/Hr
11	Blind Spot	Blind Spot	11.49390, 77.86748	10 KM/Hr
12	Turn	Medium	11.49513, 77.86656	30 KM/Hr
13	Turn	Medium	11.49525, 77.86659	30 KM/Hr

Emergency Locations

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr
Coordinates: 11.44898, 77.87410



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45348, 77.85706



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46318, 77.84913



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45697, 77.82719



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45542, 77.81725



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45631, 77.81668



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49249, 77.85821



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.49390, 77.86748



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.49513, 77.86656



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.49525, 77.86659

